DIFFERENTIAL GEOMETRY AND LIE GROUPS

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Fifth Semester

Contents

Chapter 1

MANIFOLDS

July 20th.

The primary reference for this course will be Loring Tu's *An Introduction to Manifolds.* Prerequisites include Analysis of Several Variables, Linear Algebra, and Topology. The mid-term will account for 35% of the final grade, quizzes and student lectures will account for 15%, and the final exam will account for 50%.

Simply put, a manifold is any object that is locally Euclidean.

1.1 Calculus on $\mathbb{R}^n$

A point p or element in $\mathbb{R}^n$ is associated with an n-tuple of coordinates $x^1, x^2, \ldots, x^n$. Let U be an open neighbourhood of p in $\mathbb{R}^n$. For a non-negative integer k, we say that a function $f : U \to \mathbb{R}$ is C^k*-continuous* at $p \in U$ if all partial derivatives of f of order $\leq k$ exist and are continuous at p. f is said to be C^∞ or *smooth* if f is C^k for all $k \geq 0$. f is C^k (or C^∞)on U if it is C^k (or C^∞) at every point in U.

If f happens to be a vector-valued function, i.e. $f : U \to \mathbb{R}^m$, we say $f = (f^1, f^2, \ldots, f^m)$ is C^k (or C^∞) at p if each component function $f^i : U \to \mathbb{R}$ is C^k (or C^∞) at p. f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U.

Definition 1.1. Let R be a commutative ring with unity. We term A to be an R*-algebra* if A is a commutative ring with unity and an R-module, and for $r \in R$ and $a, b \in A$, we have $r(ab) = (ra)b = a(rb)$.

In this course, we will be primarily be concerned with the case when R is a field, and A is an R-vector space.

For a subset $U \subseteq \mathbb{R}^n$, we denote $C^k(U)$ (or $C^\infty(U)$) as the set of all C^k (or C^∞) functions from U to $\mathbb{R}$. Note that this is a commutative ring with unity. It is also a vector space over $\mathbb{R}$. Together, $C^k(U)$ (or $C^\infty(U)$) is a (commutative) algebra over $\mathbb{R}$. Moreover, we have the descending chain of subalgebras

$$C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq \cdots C^k(U) \supseteq C^{k+1}(U) \supseteq \cdots \tag{1.1}$$

with

$$\bigcap_{k \geq 0} C^k(U) = C^\infty(U). \tag{1.2}$$

$f : U \to \mathbb{R}$ is termed *real analytic* at $p \in U$ if there exists a neighbourhood $V \subseteq U$ of p such that

$$f(x) = f(p) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \cdots + \frac{1}{k!} \sum_{1 \leq i_1, \ldots, i_k \leq n} \frac{\partial^k f}{\partial x^{i_1} \cdots \partial x^{i_k}}(p)(x^{i_1} - p^{i_1}) \cdots (x^{i_k} - p^{i_k}) + \cdots \tag{1.3}$$

for all $x \in V$. f is said to be real analytic on U if it is real analytic at every point in U. We denote the set of all real analytic functions on U as $C^\omega(U)$. Note that $C^\omega(U) \subseteq C^\infty(U)$.

Theorem 1.2 (*Taylor's theorem*)**.** *Let $p \in U \subseteq \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be a C^∞ function. Further assume that for all $x \in U$, the line segment from p to x is contained in U (U in this case is said to be* star-shaped *with respect to p). Then there exist $g_1, \ldots, g_n \in C^\infty(U)$ such that, for all $x \in U$,*

$$f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i), \tag{1.4}$$

with

$$g_i(p) = \frac{\partial f}{\partial x^i}(p), \quad 1 \le i \le n. \tag{1.5}$$

Proof. Let $x \in U$. Then the line segment $\{p + t(x-p) \mid t \in [0,1]\}$ is contained in U. Composing f with the map $t \mapsto p + t(x-p)$, we obtain

$$\frac{df}{dt}(p + t(x-p)) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p + t(x-p))(x^i - p^i). \tag{1.6}$$

Integrating from 0 to 1, we get

$$f(x) - f(p) = \int_0^1 \frac{df}{dt}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i)\int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i)g_i(x). \tag{1.7}$$

Clearly, because f is C^∞, each g_i is also C^∞. Further, we have $g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p)$. ∎

Let $p = (p^1, \ldots, p^n), v \in \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be C^∞ at p. The *directional derivative* of$f$ at p in the direction of v is defined as

$$\lim_{t \to 0} \frac{f(p + tv) - f(p)}{t} = \frac{d}{dt} f(p + tv)\Big|_{t=0} = \sum_{i=1}^{n} \frac{df}{dx^i}(p)v^i. \tag{1.8}$$

From here, we obtain the operator

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\Big|_p \tag{1.9}$$

Now consider all pairs of the form (f, U) where $p \in U \subseteq \mathbb{R}^n$ is an open neighbourhood of p and $f : U \to \mathbb{R}$ is C^∞. Define an equivalence relation $\sim$ on this set by declaring $(f, U) \sim (g, V)$ if and only if there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$. The equivalence class of (f, U) is denoted by $[(f, U)]$, and the set of all such equivalence classes is denoted by $C_p^\infty(\mathbb{R}^n)$. Such an equivalence class $[(f, U)]$ is called the *germ* of f at p. We can give $C_p^\infty(\mathbb{R}^n)$ the structure of a commutative ring with unity by defining

$$[(f, U)] + [(g, V)] = [(f + g, U \cap V)], \quad [(f, U)] \cdot [(g, V)] = [(fg, U \cap V)], \quad 1 = [(1, \mathbb{R}^n)]. \tag{1.10}$$

It can also be given the structure of a vector space over $\mathbb{R}$ by defining

$$\lambda[(f, U)] = [(\lambda f, U)], \quad \lambda \in \mathbb{R}. \tag{1.11}$$

Thus, $C_p^\infty(\mathbb{R}^n)$ is an $\mathbb{R}$-algebra. Now, given a germ $\alpha = [(f, U)] \in C_p^\infty(\mathbb{R}^n)$ and a vector $v \in \mathbb{R}^n$, we define

$$D_v\alpha = \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}\Big|_p = v \cdot \nabla f(p). \tag{1.12}$$

Note that the choice of f is immaterial, since if $[(f, U)] = [(g, V)]$, then there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$, and hence $\nabla f(p) = \nabla g(p)$. One can infer that $D_{\lambda v}(f) = \lambda D_v(f) = D_v(\lambda f)$ and $D_v(f + g) = D_v(f) + D_v(g)$ for all $\lambda \in \mathbb{R}$ and $f, g \in C_p^\infty(\mathbb{R}^n)$. Thus, D_v is an $\mathbb{R}$-linear map from $C_p^\infty(\mathbb{R}^n)$ to $\mathbb{R}$ (hereforth, we shall use f to also denote the germ $[(f, U)]$ of f at p). Also,

$$D_v(fg) = \sum_{i=1}^{n} v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^{n} v^i \left(f(p)\frac{\partial g}{\partial x^i}(p) + g(p)\frac{\partial f}{\partial x^i}(p) \right) = f(p)D_v(g) + g(p)D_v(f). \tag{1.13}$$

Such a property makes D_v what we call a derivation.

Definition 1.3. Any $\mathbb{R}$-linear map $T : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}$ satisfying the Leibniz rule $T(fg) = f(p)T(g) + g(p)T(f)$ for all $f, g \in C_p^\infty(\mathbb{R}^n)$ is called a *derivation* at p.

One may show that if T is a derivation at p, then $T(1) = 0$.

The set of all derivations at p is denoted by $\mathfrak{D}_p(\mathbb{R}^n)$. It is easy to see that $\mathfrak{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. Now consider the map

$$\theta : \mathbb{R}^n \to \mathfrak{D}_p(\mathbb{R}^n), \quad v \mapsto D_v. \tag{1.14}$$

This θ is an $\mathbb{R}$-linear map (verify). We claim that θ is an isomorphism of real vector spaces. To this end, let $v \in \mathbb{R}^n$ such that $D_v = 0 \in \mathfrak{D}_p(\mathbb{R}^n)$. Then $D_v f = 0$ for all $f \in C_p^\infty(\mathbb{R}^n)$. In particular, for the coordinate functions x^i, we have $D_v x^i = v_i = 0$ for all $1 \leq i \leq n$. Thus, $v = 0$, and hence θ is injective. To show surjectivity, we appeal to Taylor's theorem; let $D \in \mathfrak{D}_p(\mathbb{R}^n)$. We claim that $D = D_v$ where $v = (D(x^1), \ldots, D(x^n))$. Indeed, for any $f \in C_p^\infty(\mathbb{R}^n)$, we have

$$\begin{aligned} D(f) = D(f(p)) + D\left(\sum_{i=1}^n (x^i - p^i) g_i(x)\right) &= 0 + \sum_{i=1}^n \left((x^i - p^i)(p) D g_i(x) + D(x^i - p^i) g_i(p)\right) \\ = \sum_{i=1}^n (Dx^i) g_i(p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p) Dx^i = D_v(f). \end{aligned} \tag{1.15}$$

Hence, θ is surjective, and we have the desired isomorphism $\mathfrak{D}_p(\mathbb{R}^n) \cong \mathbb{R}^n$.

Note that a germ $f \in C_p^\infty(\mathbb{R}^n)$ has a valid value at p, but no other point. Hence, we have the evaluation map

$$\mathrm{ev}_p : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}, \quad [(f, U)] \mapsto f(p). \tag{1.16}$$

The kernel here is $\{[(f, U)] \in C_p^\infty(\mathbb{R}^n) \mid f(p) = 0\}$, which is a ideal in $C_p^\infty(\mathbb{R}^n)$. For any other element $[(f, U)] \in C_p^\infty(\mathbb{R}^n)$ with $f(p) \neq 0$, a valid inverse is given by $[(f^{-1}, U \setminus \{f = 0\})]$. This makes $C_p^\infty(\mathbb{R}^n)$ a local ring.

1.2 Multilinear Algebra

July 22nd.

Hereforth, V shall denote a finite-dimensional real vector space, and V^V shall denote its *dual space* $\mathrm{Hom}_{\mathbb{R}}(V, \mathbb{R})$. If $e_1, \ldots, e_n$ are a basis for V, then the *dual basis* of V^V is denoted by $\alpha^1, \ldots, \alpha^n \in V^V$, and is defined by $\alpha^i(e_j) = \delta_{i,j}$ for all $1 \leq i, j \leq n$. One may verify that the dual basis $\{\alpha^1, \ldots, \alpha^n\}$, induced by the basis $\{e_1, \ldots, e_n\}$, is indeed a basis for V^V. S_k shall denote the permutation group on k symbols, satisfying $|S_k| = k!$. For any $\sigma \in S_k$, the sign of σ is denoted by $\mathrm{sgn}(\sigma)$, and is defined as $+1$ if σ is an even permutation, and -1 if σ is an odd permutation.

With notation out of the way, let $k \geq 1$ be an integer. We shall consider the k-tuple $V^k = V \times \cdots \times V$ (k times).

Definition 1.4. A mapping $f : V^k \to \mathbb{R}$ is said to be *k-linear* if f is linear in each of its k arguments.

A special case is when $k = 2$, in which case f is called a *bilinear form*. Such an $f : V \times V \to \mathbb{R}$ satisfies

$$f(\lambda v + \mu w, u) = \lambda f(v, u) + \mu f(w, u), \quad f(u, \lambda v + \mu w) = \lambda f(u, v) + \mu f(u, w). \tag{1.17}$$

A common example of a bilinear form is any valid inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ on V. Another example is the determinant $\det : \mathbb{R}^n \to \mathbb{R}$, where the vectors are interpreted as the columns of a matrix. We can also write

$$L_k(V) = \{f : V^k \to \mathbb{R} \mid f \text{ is } k\text{-linear}\}. \tag{1.18}$$

It is easy to see that $L_k(V)$ is an $\mathbb{R}$-vector space.

Definition 1.5. A k-linear map $f : V^k \to \mathbb{R}$ is called a *symmetric k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.19}$$

That is, f is invariant under any permutation of its arguments. Moreover, f is called an *alternating k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \ldots, v_k) = \operatorname{sgn}(\sigma) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.20}$$

One can see that the standard inner product on $\mathbb{R}^n$ is a symmetric bilinear form, and the determinant is an alternating n-linear form. Related to these definitions, we can immediately define the following subspaces of $L_k(V)$:

$$S_k(V) = \{f \in L_k(V) \mid f \text{ is symmetric}\} \subseteq L_k(V), \quad A_k(V) = \{f \in L_k(V) \mid f \text{ is alternating}\} \subseteq L_k(V). \tag{1.21}$$

Currently, our main interest of study is $A_k(V)$, the space of alternating k-linear forms on V. Elements of $A_k(V)$ are termed *alternating k-tensors*. Similarly, those of $S_k(V)$ and $L_k(V)$ are called *symmetric k-tensors* and *k-tensors*, respectively. For $k = 0$, we define $L_0(V) = S_0(V) = A_0(V) = \mathbb{R}$. For $k = 1$, we have $L_1(V) = S_1(V) = A_1(V) = V^V$. We can define the (left) action of S_k on $L_k(V)$ by

$$(\sigma \cdot f)(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.22}$$

Thus, f is symmetric if $\sigma \cdot f = f$ for all $\sigma \in S_k$, and f is alternating if $\sigma \cdot f = \operatorname{sgn}(\sigma) f$ for all $\sigma \in S_k$.

Given an $f \in L_k(V)$, we can define the *symmetrization* $Sf \in S_k(V)$, of f, as

$$Sf = \sum_{\sigma \in S_k} \sigma \cdot f. \tag{1.23}$$

One may, indeed, check that Sf is symmetric. Similarly, we can define the *alternation* $Af \in A_k(V)$, of f, as

$$Af = \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) \sigma \cdot f. \tag{1.24}$$

1.2.1 The Tensor and Wedge Products

July 24th.

For any $f \in L_k(V)$ and $g \in L_l(V)$, we can define the *tensor product* $f \otimes g \in L_{k+l}(V)$ as

$$(f \otimes g)(v_1, \ldots, v_{k+l}) = f(v_1, \ldots, v_k) g(v_{k+1}, \ldots, v_{k+l}). \tag{1.25}$$

Verify that $f \otimes g$ is indeed $(k + l)$-linear. We are interested in the tensor product of alternating tensors, so let $f \in A_k(V)$ and $g \in A_l(V)$. Then, we can define the *wedge product* $f \wedge g \in A_{k+l}(V)$ as

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot (f \otimes g). \tag{1.26}$$

The associativity of the tensor product holds easily: for $f \in L_k(V)$, $g \in L_l(V)$, and $h \in L_m(V)$,

$$(f \otimes g) \otimes h = f \otimes (g \otimes h). \tag{1.27}$$

Proposition 1.6. *For $f \in A_k(V)$ and $g \in A_l(V)$, we have*

$$f \wedge g = (-1)^{kl} g \wedge f. \tag{1.28}$$

Proof. We shall show that $A(f \otimes g) = (-1)^{kl} A(g \otimes f)$. To this end, fix the permutation $\tau \in S_{k+l}$ defined by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ k+1 & k+2 & \cdots & k+l & 1 & 2 & \cdots & k \end{pmatrix}. \tag{1.29}$$

We have

$$
\begin{aligned}
A(f\otimes g)(v_1,\dots,v_{k+l}) &= \sum_{\sigma\in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma(1)},\dots,v_{\sigma(k)})g(v_{\sigma(k+1)},\dots,v_{\sigma(k+l)}) \\
&= \sum_{\sigma\in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma\tau(l+1)},\dots,v_{\sigma\tau(l+k)})g(v_{\sigma\tau(1)},\dots,v_{\sigma\tau(l)}) \\
&= \operatorname{sgn}(\tau)\sum_{\sigma\in S_{k+l}} \operatorname{sgn}(\sigma\tau) f(v_{\sigma\tau(l+1)},\dots,v_{\sigma\tau(l+k)})g(v_{\sigma\tau(1)},\dots,v_{\sigma\tau(l)}) \\
&= \operatorname{sgn}(\tau)A(g\otimes f)(v_1,\dots,v_{k+l}). \qquad (1.30)
\end{aligned}
$$

With $\operatorname{sgn}(\tau)=(-1)^{kl}$, the proof is complete. ∎

Corollary 1.7. *For $f\in A_k(V)$, we have $f\wedge f=0$ if k is odd.*

Proof. Easy since $f\wedge f=(-1)^{k^2}f\wedge f=-f\wedge f$ if k is odd, so $f\wedge f=0$. ∎

Lemma 1.8. *For $f\in L_k(V)$ and $g\in L_l(V)$, we have*

$$A(Af\otimes g)=k!A(f\otimes g),\quad A(f\otimes Ag)=l!A(f\otimes g). \qquad (1.31)$$

Proof. We shall prove the first equality; the second follows similarly. We have

$$
\begin{aligned}
A(Af\otimes g) &= \sum_{\sigma\in S_{k+l}}\operatorname{sgn}(\sigma)\sigma\cdot\left(\sum_{\tau\in S_k}\operatorname{sgn}(\tau)(\tau\cdot f)\otimes g\right) = \sum_{\sigma\in S_{k+l}}\operatorname{sgn}(\sigma)\sigma\cdot\left(\sum_{\tau\in S_k}\operatorname{sgn}(\tau)\tau\cdot(f\otimes g)\right) \\
&= \sum_{\tau\in S_k}\sum_{\sigma\in S_{k+l}}\operatorname{sgn}(\sigma\tau)\sigma\tau\cdot(f\otimes g) = k!\sum_{\alpha\in S_{k+l}}\operatorname{sgn}(\alpha)\alpha\cdot(f\otimes g)=k!A(f\otimes g). \qquad (1.32)
\end{aligned}
$$

∎

Proposition 1.9. *The wedge product is associative, i.e. for $f\in A_k(V)$, $g\in A_l(V)$, and $h\in A_m(V)$, we have*

$$(f\wedge g)\wedge h=f\wedge(g\wedge h). \qquad (1.33)$$

Proof. We have

$$
\begin{aligned}
(f\wedge g)\wedge h &= \frac{1}{(k+l)!m!}A((f\wedge g)\otimes h) = \frac{1}{(k+l)!m!k!l!}A(A(f\otimes g)\otimes h) = \frac{(k+l)!}{(k+l)!l!k!m!}A((f\otimes g)\otimes h) \\
&= \frac{1}{k!l!m!}A(f\otimes g\otimes h)=f\wedge(g\wedge h). \qquad (1.34)
\end{aligned}
$$

∎

Corollary 1.10. *$f\wedge g\wedge h=\frac{1}{k!l!m!}A(f\otimes g\otimes h)$ for $f\in A_k(V)$, $g\in A_l(V)$, and $h\in A_m(V)$. Inductively,*

$$f_1\wedge\cdots\wedge f_k=\frac{1}{k_1!\cdots k_k!}A(f_1\otimes\cdots\otimes f_k), \qquad (1.35)$$

where $f_i\in A_{k_i}(V)$ for $1\le i\le k$.

Proposition 1.11. *Let $\alpha^1,\dots,\alpha^n\in V^V=A_1(V)=L_1(V)$, and $v_1,\dots,v_k\in V$. Then*

$$(\alpha^1\wedge\cdots\wedge\alpha^k)(v_1,\dots,v_k)=\det(\alpha^i(v_j))_{1\le i,j\le k}. \qquad (1.36)$$

Proof. We have

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) = \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)(\alpha^1 \otimes \cdots \otimes \alpha^k)(v_{\sigma(1)}, \ldots, v_{\sigma(k)})$$
$$= \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\alpha^1(v_{\sigma(1)})\alpha^2(v\sigma(2)) \cdots \alpha^k(v_{\sigma(k)}) = \det(\alpha^i(v_j))_{1 \leq i,j \leq k}. \tag{1.37}$$

■

July 27th.

Let V be an $\mathbb{R}$-vector space with basis $e_1, \ldots, e_n$, and V^V be its dual space with dual basis $\alpha^1, \ldots, \alpha^n$. Moreover, let I be a multi-index of length k, i.e. $I = (i_1, \ldots, i_k)$ with $1 \leq i_1, \ldots, i_k \leq n$. We define $e_I := (e_{i_1}, \ldots, e_{i_k}) \in V^k$ and $\alpha^I := \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} \in A_k(V)$. Then, a k-linear form $f \in L_k(V)$ is completely determined by its values on the e_I, and an alternating k-linear form $f \in A_k(V)$ is completely determined by its values on the e_I with $i_1 < i_2 < \cdots < i_k$.

To this end, we first claim that $\alpha^I(e_J) = \delta^I_J$ for all multi-indices I, J of length k. Indeed, if $J = (j_1, \ldots, j_k)$ with $j_1 < j_2 < \cdots < j_k$, then

$$\alpha^I(e_J) = \det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k}; \tag{1.38}$$

. this is 1 for $I = J$. Suppose $I \neq J$. Let $1 \leq l \leq k$ be the first index where $i_l \neq j_l$. Then the l^{th} row of $\alpha^{i_m}(e_{j_n})_{1 \leq m,n \leq k}$ is an all zeroes row, and hence $\det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k} = 0$. Thus, $\alpha^I(e_J) = \delta^I_J$. We now claim that α^I, for strictly increasing multi-indices I of length k, form a basis for $A_k(V)$. For linear independence, we simply have (the sums in this context are over strictly increasing multi-indices I of length k):

$$\sum_I c_I \alpha^I = 0 \implies \sum_I c_I \alpha^I(e_J) = 0 \implies c_J = 0, \quad \forall J. \tag{1.39}$$

To show that they span $A_k(V)$, let $f \in A_k(V)$. Then consider the k-linear form

$$g = \sum_I f(e_I)\alpha^I. \tag{1.40}$$

We show that $f = g$; it is enough to prove that $f(e_J) = g(e_J)$ for all strictly increasing multi-indices J of length k. We have

$$g(e_J) = \sum_I f(e_I)\alpha^I(e_J) = f(e_J)\alpha^J(e_J) + \sum_{I \neq J} f(e_I)\alpha^I(e_J) = f(e_J). \tag{1.41}$$

Corollary 1.12. *$A_k(V) = 0$ if $k > n$. Further,* $\dim A_k(V) = \binom{n}{k}$ *if $1 \leq k \leq n$.*

1.3 Building Blocks for Manifolds

Recall that we had defined the class of germs $C^\infty_p(\mathbb{R}^n)$ of C^∞ functions at a point $p \in \mathbb{R}^n$, which agree on some neighbourhood of p. One can show that $C^\infty_p(\mathbb{R}^n)$ and $C^\infty_p(U)$ are isomorphic as $\mathbb{R}$-algebras, where $U \subseteq \mathbb{R}^n$ is an open neighbourhood of p. Note that we term $T_p(U) = T_p(\mathbb{R}^n) = \mathbb{R}^n$ as the *tangent space* of U (or $\mathbb{R}$) at p.

Definition 1.13. Let $U \subseteq \mathbb{R}^n$ be an open subset. A *vector field* on U is a function $X : U \to \amalg_{p \in U} T_p(\mathbb{R}^n)$ such that $X(p) \in T_p(\mathbb{R}^n)$ for all $p \in U$. In particular,

$$X(p) = \sum_{i=1}^n a^i(p) \frac{\partial}{\partial x^i}\bigg|_p, \quad a^i(p) \in \mathbb{R}. \tag{1.42}$$

We can rewrite this as

$$X = \sum_{i=1}^{n} a^i \frac{\partial}{\partial x^i}, \quad a^i : U \to \mathbb{R}. \tag{1.43}$$

X is called a *smooth vector field* if each a^i is a smooth function on U. The set of all smooth vector fields on U is denoted by $\mathfrak{X}(U)$.

This is a real vector space; note that for $X, Y \in \mathfrak{X}(U)$, we have $X + Y \in \mathfrak{X}(U)$, and for $\lambda \in \mathbb{R}$, we have $\lambda X \in \mathfrak{X}(U)$. Further, $\mathfrak{X}(U)$ is a $C^\infty(U)$-module since for $f \in C^\infty(U)$ and $X \in \mathfrak{X}(U)$, we have $fX \in \mathfrak{X}(U)$, where

$$(fX)(p) = f(p)X(p), \quad p \in U. \tag{1.44}$$

Notice that $X \in \mathfrak{X}(U)$ describes a map $C^\infty(U) \to C^\infty(U)$, where $f \mapsto Xf$, being defined as $(Xf)(p) = X_p f$. This is a derivation on $C^\infty(U)$, since for $f, g \in C^\infty(U)$, we have

$$(X(fg))(p) = X_p(fg) = f(p)X_p g + g(p)X_p f = (fXg)(p) + (gXf)(p). \tag{1.45}$$

We refine our definition of a derivation as follows.

Definition 1.14. Let K be a field, and A be a K-algebra. A *derivation* $D : A \to A$ is an $\mathbb{R}$-linear map satisfying the Leibniz rule $D(ab) = aD(b) + bD(a)$ for all $a, b \in A$.

Definition 1.15. A *differential 1-form* on U is a smooth function $\omega : U \to \amalg_{p \in U} T_p(\mathbb{R}^n)^V$ such that $\omega(p) \in T_p(\mathbb{R}^n)^V =: T_p^*(\mathbb{R}^n)$ for all $p \in U$.

For example, for $f \in C^\infty(U)$, we can define the differential 1-form df on U by $df(p) = df_p \in \mathrm{Hom}_{\mathbb{R}}(T_p\mathbb{R}^n, \mathbb{R})$, where $df_p(X) = X_p f$ for all $X \in T_p\mathbb{R}^n$. In particular, dx^i is a differential 1-form on U for all $1 \leq i \leq n$. In fact, $\{dx^1, \ldots, dx^n\}$ is the dual basis of $T_p^*\mathbb{R}^n$ corresponding to the basis $\{\frac{\partial}{\partial x^1}\Big|_p, \ldots, \frac{\partial}{\partial x^n}\Big|_p\}$ of $T_p\mathbb{R}^n$. Thus, a differential 1-form ω on U can be expressed as

$$\omega(p) = \sum_{i=1}^{n} a_i(p) dx^i\Big|_p, \quad a_i : U \to \mathbb{R}. \tag{1.46}$$

$T_p^*(\mathbb{R}^n)$, here, is termed the *cotangent space* of $\mathbb{R}^n$ at p. Thus, our current setup is as follows: $V = T_p\mathbb{R}^n$, and $V^V = T_p^*\mathbb{R}^n$. The basis for $T_p\mathbb{R}^n$ considered is $\{\frac{\partial}{\partial x^1}\Big|_p, \ldots, \frac{\partial}{\partial x^n}\Big|_p\}$, and the corresponding dual basis for $T_p^*\mathbb{R}^n$ is $\{dx^1\Big|_p, \ldots, dx^n\Big|_p\}$. And finally, we have

$$A_k(T_p\mathbb{R}^n) = \text{the space of alternating } k\text{-linear forms on } T_p\mathbb{R}^n \tag{1.47}$$

which has the basis $\{dx^{i_1}\Big|_p \wedge \cdots \wedge dx^{i_k}\Big|_p \mid 1 \leq i_1 < \cdots < i_k \leq n\}$.

Definition 1.16. A *differential k-form* on U is a smooth function $\omega : U \to \amalg_{p \in U} A_k(T_p\mathbb{R}^n)$ such that $\omega(p) \in A_k(T_p\mathbb{R}^n)$ for all $p \in U$.

Any such ω may be written as $\omega = \sum_I a_I dx_I$. It termed a smooth differential k-form if each a_I is a smooth function on U. The set of all smooth differential 1-forms on U is denoted by $\Omega^1(U)$, and the set of all smooth differential k-forms on U is denoted by $\Omega^k(U)$. Both these sets are $\mathbb{R}$-vector spaces, and $\Omega^k(U)$ is a free $C^\infty(U)$-module of rank $\binom{n}{k}$. The isomorphism may be given as

$$\Omega^1(U) \cong C^\infty(U)^{\oplus \binom{n}{k}}, \quad \omega = \sum_I a_I dx_I \mapsto (a_1, a_2, \ldots, a_{\binom{n}{k}}). \tag{1.48}$$

We can endow $\Omega^k(U)$ with the wedge product $\wedge : \Omega^k(U) \times \Omega^l(U) \to \Omega^{k+l}(U)$, defined as

$$(\omega \wedge \eta)(p) = \omega(p) \wedge \eta(p), \quad p \in U, \tag{1.49}$$

or

$$\omega \wedge \eta = \sum_{I,J} a_I b_J dx_I \wedge dx_J \tag{1.50}$$

where I and J are strictly increasing multi-indices of length k and l, respectively. Note that if $I \cap J \neq \emptyset$, then $dx_I \wedge dx_J = 0$. Thus, the sum may be taken over all strictly increasing multi-indices I and J of length k and l, respectively, such that $I \cap J = \emptyset$. Also,

$$A_*(V) = \bigoplus_{k=0}^{n} A_k(V) \tag{1.51}$$

is defined as the *exterior algebra* of V. The wedge product on here is associative and anti-commutative. We define $\Omega^0(U) = C^\infty(U)$. The map $d : \Omega^k(U) \to \Omega^{k+1}(U)$ is defined as the *exterior derivative*; for $k = 0$, we have

$$d : C^\infty(U) \to \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i. \tag{1.52}$$

For a k-form $\omega = \sum_I a_I dx_I \in \Omega^k(U)$, we have

$$d\omega = \sum_I da_I \wedge dx_I = \sum_I \sum_{j=1}^{n} \frac{\partial a_I}{\partial x^j} dx^j \wedge dx_I. \tag{1.53}$$

Example 1.17. Let $U \subseteq \mathbb{R}^3$. Let us call the coordinates $x^1 = x$, $x^2 = y$, and $x^3 = z$. The only non-zero forms are $\Omega^0(U)$, $\Omega^1(U)$, $\Omega^2(U)$, and $\Omega^3(U)$. Note that $\Omega^0(U) = C^\infty(U)$, and there's nothing to say about this. For $\Omega^1(U)$, a form looks like $f\,dx + g\,dy + h\,dz$ for $f, g, h \in C^\infty(U)$. For $\Omega^2(U)$, a form looks like $R\,dx \wedge dy + Q\,dx \wedge dz + P\,dy \wedge dz$ for $f, g, h \in C^\infty(U)$.

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